Kaan: On-Device Acoustic Screening for Stored-Grain Pests under Phone-Mic Domain Shift

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Abstract

Post-harvest insect damage is a major food-security and climate-relevant loss pathway in stored grain. We present **Kaan**, an open Apache-2.0 smartphone system that classifies contact acoustics into clean grain versus three primary stored-rice pests with a compact INT8 CNN (TFLite / ONNX), offline inference, and multilingual advisories. We emphasize deployment constraints over peak lab accuracy alone: leakage-aware file-level evaluation after byte-dedupe, an eleven-approach bake-off with multi-seed statistics, teacher distillation into a phone-sized student, temperature calibration with abstention, SimCLR-style mel SSL, cost-sensitive and strict-gated weevil↔borer heads, and a *robustness ladder* of phone-like degradations without field recordings. A distilled production model reaches 97.15% validation accuracy; multi-seed baseline accuracy is $96.73\% \pm 0.37\%$ (seeds 42/43/44). Classical and deep models both exceed a soft 84.51% literature reference under our protocol. The robustness ladder shows mild muffling remains strong while telephone band-limiting and additive noise collapse accuracy (phone-band $60.9\% \pm 8.4\%$; SNR$\leq 10\,$dB / hard combos $\sim 7.6\%$), a first-class finding for rural handset deployment.

1 Climate impact and deployment goal

Reducing avoidable post-harvest loss improves food availability and lowers pressure to expand production and emissions IGMRI (2015); World Bank and FAO. Internal feeders are hard to see early; acoustic screening can trigger aeration, sunning, or fumigation sooner than visual inspection alone. Kaan targets *adaptation* (protecting stores under climate stress) and indirect *mitigation* (less waste), optimized for offline phones: no cloud inference, TFLite/ONNX footprints, and advisories in English, Hindi, Marathi, Punjabi, and Telugu. It is a screening aid, not a laboratory or legal diagnosis.

2 Problem, data, and system

Classes: clean; rice weevil (*Sitophilus oryzae*); lesser grain borer (*Rhyzopertha dominica*); red flour beetle (*Tribolium castaneum*). **Pipeline:** $10\,$s @ $16\,$kHz mono $\rightarrow$ trim/pad $\rightarrow$ mel $128 \times 128 \times 1$ (n_{fft}=2048, hop 512, 128 mels, dB, min-max) $\rightarrow$ INT8 CNN $\rightarrow$ confidence gate (0.6). Pest audio is from the IRRI Rice Acoustic Sensor dataset Balingbing et al. (2024); clean uses Speech Commands ambient windows (not synthetic silence). Splits are stratified at *file* level after byte-deduplication. Shipped artifacts: Keras H5, INT8 TFLite ($\sim$333 KB), ONNX for browser inference.

3 Methods

Bake-off. Eleven approaches share one leakage-aware split: mel-CNNs (shallow/deep), 1D CNN, YAMNet probe, and classical models on $\sim$74-D handcrafted features (SVM-RBF, MLP, GBDT, RF,

Table 1: Multi-seed bake-off accuracy (mean±std; seeds 42/43/44). Soft reference: cited 84.51% Balingbing et al. (2024) under *our* protocol.

Approach	Acc mean±std	Seeds > 84.51%
GBDT	95.36 ± 1.50	3/3
CNN deep	95.15 ± 0.48	3/3
ExtraTrees	94.94 ± 1.10	3/3
SVM-RBF	94.73 ± 1.20	3/3
YAMNet probe	85.65 ± 1.83	2/3
CNN-1D	64.77 ± 20.7	1/3

ExtraTrees, kNN, logistic regression). We report multi-seed means (42/43/44), bootstrap CIs, and McNemar where relevant.

Distillation. Soft labels from GBDT + ExtraTrees + deep CNN train a deep student for production export (TFLite + ONNX).

Robustness ladder. Replay validation audio under pink SNR, telephone band-pass (300–3400 Hz), muffling, compression, clipping, reverb, gain, and combinations; report accuracy per rung.

Calibration. Temperature scaling on validation; ECE/NLL before vs after; abstain curves trade coverage for accuracy.

Heads. Cost-sensitive weevil↔borer upweighting; fine-tuned pair specialist on the baseline encoder with *strict* fusion (override only when baseline top-2 are exactly that pair and margin is below a validation-swept gate).

SSL. SimCLR-style SpecAugment views on IRRI+ambient mels, then supervised fine-tune.

4 Results

Bake-off (soft reference 84.51%). Table 1: GBDT and deep CNN both clear the reference on all three seeds; classical ≈ deep under this protocol; main confusion is weevil↔borer; CNN-1D is unstable.

Distillation (seed 42). Teacher ensemble 95.89%; hard-only deep 95.57%; **distilled student** 97.15% (macro-F1 0.977), shipped as production weights.

Advanced suite (seeds 42/43/44; bootstrap 95% CI of the mean). Baseline accuracy $96.73\% \pm 0.37\%$ $[96.52, 97.15]$; SSL $96.62\% \pm 0.18\%$; cost-sensitive $96.73\% \pm 0.66\%$; hierarchical $94.09\% \pm 4.76\%$ (higher variance). ECE $0.042 \rightarrow 0.029$ after temperature scaling (mean $T{\approx}0.67$). Strict-gated hierarchical fine-tune on seed 42: baseline $96.84\% \rightarrow$ fused 97.15% (pair subset 0.96; scratch cascade 94.30%).

Robustness (Table 2). Mild muffling/reverb/gain remain high; phone band-pass and noise collapse. Laboratory accuracy does **not** imply phone-mic field accuracy.

5 Limitations

No Indian phone-on-bag labeled corpus yet (robustness ladder is a proxy). Only three pest species. The 84.51% comparison is soft (not a locked reimplementation). Source data are limited in geography and grain type. Screening aid only: abstain when confidence is low.

Table 2: Robustness ladder (3-seed mean±std on baseline).

Rung	Acc mean±std
Clean	$96.73 \pm 0.37\%$
Gain / reverb / muffle	~ 91–97%
Compress	$87.87 \pm 2.85\%$
Clip	$77.85 \pm 9.66\%$
Phone band-pass	$60.86 \pm 8.35\%$
SNR 20 dB	$16.03 \pm 10.90\%$
SNR ≤ 10 dB / hard combos	$\sim 7.59\%$

6 Open release

Apache-2.0 code, distilled weights, bake-off and advanced-suite JSON/Markdown (including multi-seed CIs), hierarchical fine-tune reports, and a static results dashboard are public.[1] Next: labeled phone-on-bag collection with extension partners and phone-matched augmentation.

References

C. Balingbing et al. Application of a multi-layer CNN and acoustic device for stored rice pests. *Computers and Electronics in Agriculture*, 225:109297, 2024.

IGMRI. Annual Report. Indian Grain Storage Management & Research Institute, 2015.

World Bank / FAO reports on post-harvest loss (food loss & waste). See also regional grain storage extension literature.

[1] Anonymized for review; camera-ready will cite the repository.